Financial Triggers: A Multi-dimensional Data Mining Analysis of Mental Health in NYC

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***Abstract*—Mental health disorders in urban populations are increasingly influenced by environmental stressors such as housing instability, chronic illness, and limited access to healthcare. This study investigates the relationship between psychological distress and real-world health outcomes in New York City using a multi-dimensional data mining approach. We integrate three key datasets: (1) a BRFSS-derived public health survey containing behavioral and demographic indicators, (2) New York State hospital discharge data filtered for mental health diagnoses, and (3) ZIP-level income data from the U.S. Census.**

Machine learning classification models, including Logistic Regression, Random Forest, and XGBoost, were used to predict mental health risk, with the Kessler Psychological Distress Scale (K6) guiding the creation of the target variable. Logistic Regression achieved the highest F1 score of 0.66, identifying rent payment delays and unemployment as primary predictors of psychological distress. Cross-dataset regression analysis revealed a statistically significant correlation between regional income and hospital admissions for mental illness, though income did not predict the length of hospital stays.

These results highlight how behavioral and clinical indicators can be integrated through data mining to support public health decision-making. The findings may guide targeted intervention strategies and improve mental health service delivery across New York City communities.

I. Introduction

Mental health disorders ranging from depression and anxiety to substance abuse continue to pose serious challenges for public health systems, particularly in densely populated and socioeconomically diverse urban centers like New York City. These conditions are not only medical but are also shaped by environmental and economic triggers, including housing instability, job insecurity, lack of education, and income inequality.

Over the last decade, the mental health crisis in NYC has deepened, particularly among low-income residents. Individuals experiencing financial hardship are more likely to face stress-related disorders due to compounding factors such as rent burden, unemployment, and limited access to care. With increasing inflation and rising living costs, understanding how economic burden affects mental health outcomes has become more important than ever.

In this project, we explore how financial and environmental factors contribute to mental health risks across New York City using data mining and machine learning techniques. Our approach combines multiple datasets, including a public health survey (capturing mood, behavioral, and socioeconomic traits), hospitalization data (with over 2 million mental health-related records), and income data at the ZIP-code level. These datasets, when analyzed together, offer a comprehensive view of how economic status and regional disparities relate to mental health.

II. Project Objectives

The primary objective of this project is to explore the intersection between financial conditions and mental health outcomes in New York City using data-driven approaches. To achieve this, we define the following technical and analytical goals:

- **Predict Mental Health Risk at the Individual Level**
  Apply supervised machine learning techniques (e.g., Logistic Regression, Random Forest, XGBoost) to classify individuals as either at-risk or not at-risk for mental health disorders based on behavioral, demographic, and socioeconomic features in the survey dataset. Important predictive features include mood indicators, alcohol use, rent payment delays, and income brackets.
- **Assess Regional Disparities in Mental Health Hospitalizations**
  Merge ZIP-level income data with hospital admission records to analyze how financial status across neighborhoods correlates with mental health-related hospitalizations. This objective aims to identify ZIP3 regions with disproportionately high mental health burdens and highlight systemic disparities.
- **Evaluate the Impact of Income on Length of Stay in Mental Health Facilities**
  Perform ordered logistic regression to assess whether income level significantly influences the duration of hospital stays for mental health conditions. The goal is to determine if economic status affects not just access to mental health services but also treatment intensity or continuity.

We applied machine learning classification models (Logistic Regression, Random Forest, XGBoost) to predict mental health risk, and regression analysis to explore income effects on hospitalization trends. Our findings aim to inform public policy and promote better resource allocation in communities most in need of mental health services.

III. Literature Review

The relationship between financial status and mental health has been extensively studied in public health and social science research. Economic stressors such as low income, housing instability, unemployment, and lack of access to healthcare have been shown to significantly increase the risk of mental health conditions such as depression and anxiety [1]. These risk factors are particularly relevant in urban environments like New York City, where income inequality and regional disparities in resource access are prevalent.

Nationally, the Behavioral Risk Factor Surveillance System (BRFSS) has been a valuable resource in understanding individual-level predictors of mental health. Zahran et al. [2] demonstrated how BRFSS indicators such as poor self-reported mental health days, alcohol use, and physical inactivity are reliable proxies for assessing community mental wellness. Barrett and Scott [3] further showed that when combined with machine learning methods, BRFSS data can effectively predict at-risk populations and guide early interventions.

Hospital discharge data also provides insights into population-level mental health trends. Andersen and Newman [4] found strong correlations between ZIP-code level income and rates of psychiatric hospitalization, including readmission rates and length of stay. Their findings emphasize the importance of linking economic and clinical data to understand systemic disparities in mental healthcare access and quality.

Recent studies have incorporated machine learning to identify mental health risks based on a mix of behavioral, demographic, and economic indicators. Nguyen et al. [5] used logistic regression and random forest models to predict psychological distress among university students using survey features like financial stress, sleep patterns, and academic load. Similarly, Zhang and Liu [6] proposed a multi-source fusion model that combines clinical data with social factors to improve depression risk detection.

Beyond individual-level modeling, spatial analysis techniques have been applied to detect geographic trends in mental health outcomes. Besbris and Faber [7] used geospatial clustering to identify urban mental health hotspots driven by neighborhood-level inequalities, supporting the idea that mental illness is deeply linked to environmental context.

While these studies contribute valuable insights, few have attempted to integrate survey, clinical, and income data simultaneously. This project builds on prior work by combining these dimensions to develop a data-driven framework for analyzing mental health risks and socioeconomic disparities across New York City ZIP3 regions.

IV. Methodology

A. Data Collection

The Survey dataset serves the foundation for building classification to predict who is at mental health risk that where collected from Behavioral Risk Factor Surveillance System(BRFSS) that contains individual-level features such as mood, alcohol use, education, BMI, and socioeconomic variables. Hospital dataset is from NY state Department of Health (2020) dataset that contains anonymized inpatient discharge records throughout New York State for the year of 2020.The main concern here were ZIP code information(the first three digits) were only included might be for patient privacy. The dataset had included both primary and secondary diagnosis codes but for the purpose of this analysis, mental health disorder such as depression, anxiety, bipolar disorder, schizophrenia spectrum disorders and substance use disorders were only looked. This ensures a focused examination of mental health hospitalizations across New York State, focusing on strict confidentiality standards. Income dataset were sources from U.S. Census American Community Survey(ACS) and New York State Open Data portals, with ZIP-level average household income information across New York State. This data would be used to contextualize the spatial and socioeconomic distribution of mental health risk and hospitalizations with analysis of how income disparities may relate to patterns of inpatient admissions for mental health condition. In conclusion, integrating ZIP-level income data to mental health hospitalization records would be looked in fact to meet overall study objective.

B. Data Preprocessing

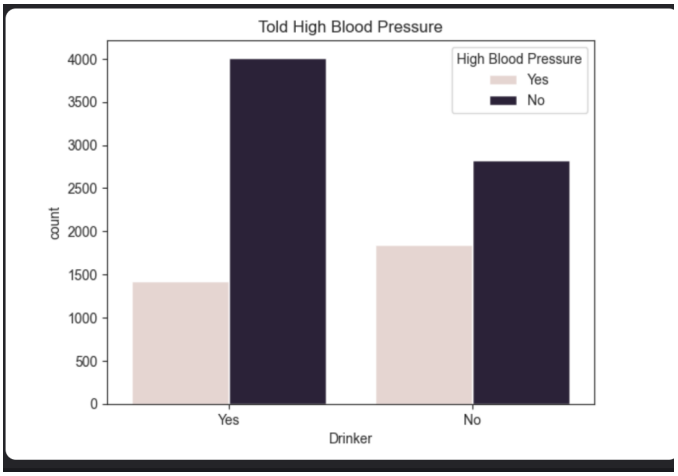
Survey Dataset Preprocessing: During this dataset preprocessing, to handle the categorical fields of attributes such as income group(imputed_povgroup3), alcohol use(drinker), education, mood were processed through label-encoded into numeric form. As, Label Encoding were assigned as integer value to each of the categorical value in column of the dataset to prepares for modeling, save memory space and used when categories have an order. With this dataset, Encode income groups were label encoding with 1,2, 3(Low, Middle, High) with created target variable **at_risk** from mood scores of 1 = at_risk and 0= not at_risk. In context of dataset, Alcohol_Use with values:['Yes', 'No'] it goit converted with label endoing to ['Yes'- 1, 'No'- 0].

In addition, numerical columns such as BMI and income were standardized to make sure that it remains consistency in scale across features. Missing values were addresses with conditional strategies where missing values in low-volume fields were removed and essential fields with missing data were imputed with statistical methods like mean or mode. Target variable were created along mood indicators and mental health responses, with labels like "at-risk" or "not at-risk" for mental health concerns.

K6 Psychological Distress Scale: The target variable for mental health risk in this study was informed using the Kessler Psychological Distress Scale (K6), a clinically validated six-question screening tool designed to assess non-specific psychological distress in large population surveys. The BRFSS dataset incorporates K6-based questions to evaluate individual mental health conditions. We used these scores to help determine whether a respondent should be labeled as "at risk" or

"not at risk" of mental health disorders, which significantly improved the robustness of our classification target.

Additionally, as part of preprocessing, we explored physical health indicators that may correlate with psychological distress. One such example is the relationship between self-reported high blood pressure and alcohol consumption. Our analysis showed that among both drinkers and non-drinkers, the majority had not been told they have high blood pressure. However, a higher proportion of non-drinkers were aware of having high blood pressure, which could suggest increased health consciousness or more frequent medical screenings. Insights like this helped us better understand lifestyle contributors to distress and refine feature selection for modeling.



Hospital Admission Dataset Preprocessing: For this dataset we imported millions of inpatient records from New York State for 2020 with range of medical admissions. And to focus on mental health related cases for the study, we filtered the data using ICD-10 diagnostic codes(F01-F99)- this stands for the international Classification of Diseases used to code and classify diseases, symptoms, injuries and other health-related conditions example code F32.0 for describing Major depressive disorder, single episode, mild or F41.1 for generalized anxiety disorder or f31.9 for Bipolar disorder, unspecified and so on. May be with privacy concern, only the first three digits of ZIP code were retained, enabling regional analysis while protecting patient anonymity. For spatial pattern analysis, we measured the frequency of mental health admissions by these ZIP codes. Additionally, for further cleaning we removed records with missing or invalid ZIP codes, and eliminated duplicate entries also date columns were converted to datetime format with features such as pear and month were extracted to support temporal trend analysis. The dataset was clean, organized regionally with collection of mental health hospitalization and ready for geographic visualization and analyze socioeconomic data to explore disparities across New York State.

Income dataset Preprocessing:: For the income dataset, we began by loading ZIP code-level economic data for New York State focusing on key columns such as ZIP code, Average Household Income and population. The plan was to merge with hospital dataset, so to ensure consistency, ZIP codes were converted to strings and converted to the first three digits allowing for regional alignment and privacy. Any non-numeric or out-of-state ZIP codes were removed, and missing values in essential columns like average income were addressed—either by dropping rows with few missing values or, if necessary, imputing with the median income of nearby ZIPs. Outlier detection was performed to flag extremely high-income ZIP codes, which were retained to highlight regional income disparities. The data were aggregated by ZIP3 for calculating the mean average income for each region to match the structure of hospital dataset. Now, the resulting dataset was clean, and ready to be merged with mental health hospitalization data for comparative analysis and visualization.

C. Exploratory Data Analysis (EDA)

EDA was performed to uncover underlying patterns or relationships within these three dataset to visualize and understand how behavioral, emotional or socioeconomic factors such as mood, alcohol use, BMI, income, education and so on in relate to self-report mental health status to identify potential predictors and creation of meaningful risk label. With hospital admission dataset, When applied to the hospital admission dataset, EDA helped isolate mental health–related cases and revealed how psychiatric hospitalizations are distributed across regions, age groups, and genders, highlighting possible disparities and providing a real-world clinical perspective to complement self-reported survey data. Similarly, for income dataset, economic landscape across New York ZIP codes were clarified. Altogether, EDA here helps in understanding the structure if data, spot trends and patterns, detect relationships with uncover anomalies provides the foundational insights of necessary for feature selection and check assumptions.

Survey Dataset EDA(processed_survey_dataset.ipynb): The cleaned survey dataset consists of 2350 rows and 15 columns in respect to behavioral, emotional and socioeconomic indictors.

1) Mood Levels by Income Group:: This visualization provides the relation between income level and average mood scores reported through survey dataset. The dataset was grouped into three income groups- Low Income, Middle Income and High Income using a label encoding with categorical column named imputed_povgroup3. Average mood level was calculated from mood responses along with respondent count to understand how many individuals contributed to each group.

	imputed_povgroup3	avg_mood_level	respondent_count	income_group
0	1	4.209358	3783	Low Income
1	2	4.363340	1533	Middle Income
2	3	4.449351	3465	High Income

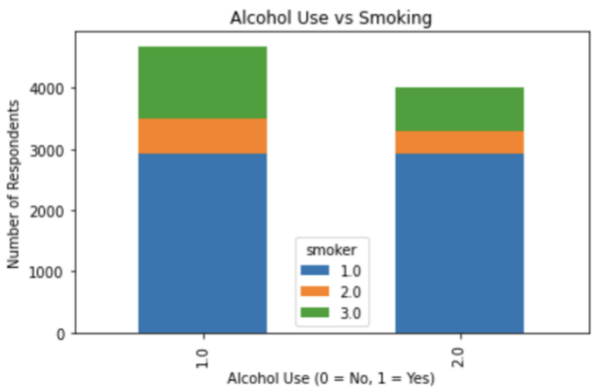
Here, income_group is income category label, avg_mood_level is the average mood score for the group and respondent_count is number of survey participants in regard

to income category. From the graph, we see Low-income individuals reported a lower average mode of 4.21, while middle Income of 4.36 and high income individuals with higher average mood of 4.45.

Alcohol use vs Smoking Status Among Respondents

This graph shows the relationship between alcohol use and smoking behavior using crosstab of the two behavioral variables Alcohol Use and Smoker levels.

```
Crosstab: Alcohol Use vs Smoking
 smoker    1.0  2.0   3.0
drinker
1.0       2925  567  1195
2.0       2923  364   719
```



Hospital Admission Dataset (hospital_admission_analysis_2020.ipynb)

The original dataset consists of 2052408 rows and 33 columns of inpatient discharge records from New York State in 2020. Only to focus the analysis on mental health related cases, we examined the "APR MDC Description" column and filtered records along keywords such as "MENTAL" specific categories like "MENTAL DISEASES AND DISORDERS" and "ALCOHOL/DRUG INDUCED ORGANIC MENTAL DISORDERS". After this, the refined dataset contained 1957233 rows and 33 columns, specifically representing mental illness admissions. This cleaned dataset would be used for Exploratory Data Analysis(EDA) to uncover several trends.

(i) Sample Records from Hospital Admission Dataset

The table shows Hospital Service Area to describe Geographic region covered, Hospital Country- Country where the hospital is located, Permanent Facility Id- unique hospital identifiers, Facility Name- Name of the medical facility, Age Group- Patient's age range, Zip Code-3 digits- the first 3 digits of hospital's ZIP code and so on.

	Hospital Service Area	Hospital County	Operating Certificate Number	Permanent Facility Id	Facility Name	Age Group	Zip Code - 3 digits	Gender	Race	Ethnicity	...	APR Risk of Mortality	APR Medical Surgical Description	Payment Typology 1	Payment Typology 2
0	New York City	Bronx	7000006.0	3058.0	Montefiore Med Center - Jack D Weiler Hosp of ...	50 to 69	104	F	Black/African American	Not Span/Hispanic	...	Moderate	Medical	Medicare	Nah
1	New York City	Bronx	7000006.0	1168.0	Montefiore Medical Center-Wakefield Hospital	18 to 29	104	F	Black/African American	Not Span/Hispanic	...	Minor	Medical	Medicaid	Nah
2	New York City	Bronx	7000006.0	1168.0	Montefiore Medical Center-Wakefield Hospital	50 to 69	104	F	Other Race	Spanish/Hispanic	...	Major	Medical	Medicare	Medicaid
3	New York City	Bronx	7000006.0	1169.0	Montefiore Medical Center - Henry & Lucy Moses...	0 to 17	104	M	Other Race	Spanish/Hispanic	...	Moderate	Medical	Medicaid	Nah
	New				Montefiore Medical Center -	0 to								Blue	

(ii) Filtering and Validating ZIP Code Prefixes in Hospital Admission Data

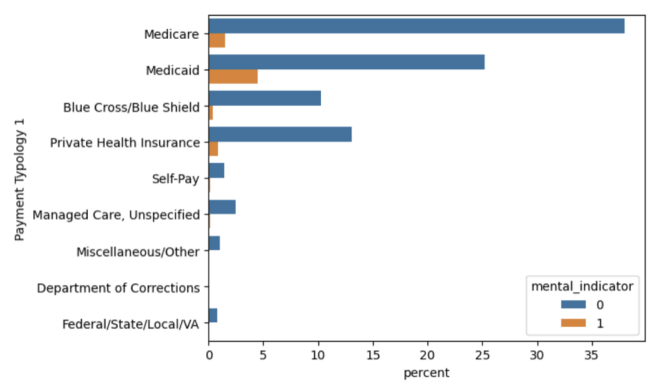
```
df_original['Zip Code - 3 digits'].unique()
```

```
array(['104', '114', '107', '100', '105', '111', '106', '108', '112',
       '125', '113', '109', '117', '127', '101', '103', '110', '115',
       '119', '121', '116', '138', '137', '124', '134', '120', '118',
       '126', '146', '122', '123', '148', '136', '131', '130', '128',
       '133', '139', '142', '140', '141', '147', '145', '144', '143',
       '149', '132', '135', '129'], dtype=object)
```

Conversion of ZIP code values to String format for easier filtering and consistency also filters out invalid ZIP codes, it displays all unique 3-digit ZIP code prefixes and prepares data for regional aggregation.

(iii) How people afford Health care:

The visualization shows hoe people pay their healthcare that connects economic access to mental health outcomes.

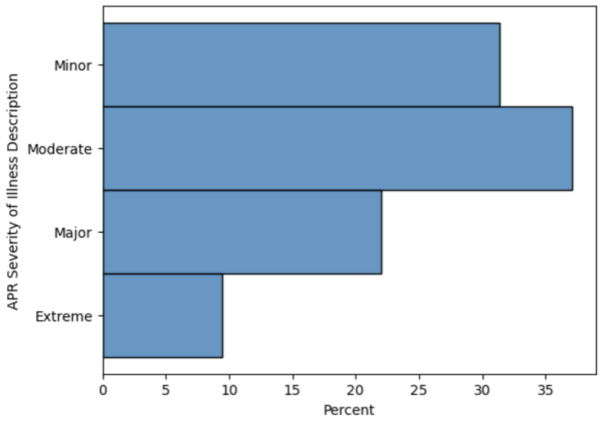


Here Y-axis indicates payment Typology like Medicare, Medicaid, Private Insurance and so on and X-axis represents percentage of total patients using each payment method. Blue color(mental_indicator-0) shows patient without a mental health diagnosis. Orange(mental_indicator-1) are patient with a mental health condition. We could see Medicare and Medicaid is most common payment methods particularly among patient with mental health conditions.

As Medicare is a government program for low-income individuals and Medicare covers elderly or disable individuals, this could suggest that individuals often belong to economically disadvantaged groups. We could insights like the low income

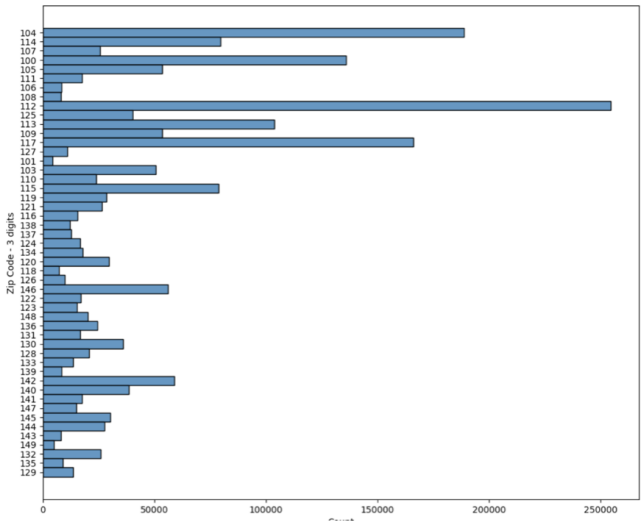
and unstable access to resource can increase mental health vulnerability.

(iv) Distribution of Hospitalized Patients by Illness Severity Level:



Y-axis here shows levels of illness severity(Minor, Moderate, Major or Extreme and X-axis shows the percentage of total hospital records. The histplot shows what percentage of patient fall into each severity category. Moderate approximately 35% most common severity, Minor appx. 32%, Major around 23%- more serious cases requiring more resources and Extreme around 10 % -might be life0threatening cases. This helps us understand the intensity and resource demand of hospitalizations.

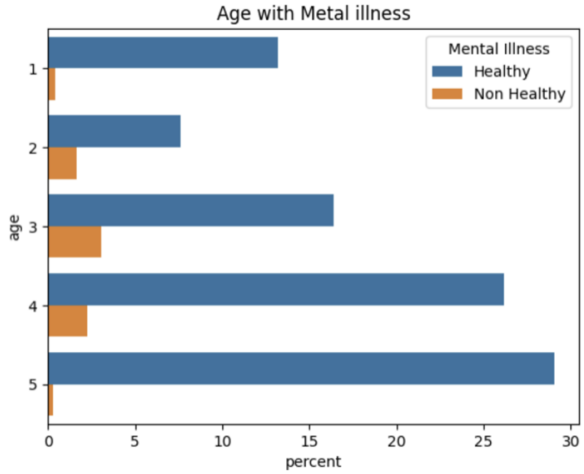
(v) Hospital admissions per ZIP Code prefix(3-gitis ZIP)



This shows the number of hospital admissions per ZIP code prefix along Zip code of 3 digits. Y-axis shows the ZIP code(3-digit prefixes like 104, 112 and so on) and X-axis shows number of hospitalization records for each ZIP3 and the chart is sorted by ZIP code. Here from graph we could see ZIP Code 112 has the highest number of hospitalization likely, Brooklyn.

This helps us identify geographic hotspots for hospitalizations which could reflects higher healthcare demand and when paired with income data, we could reveal whether low-income ZIPs experience more hospitalizations or not.

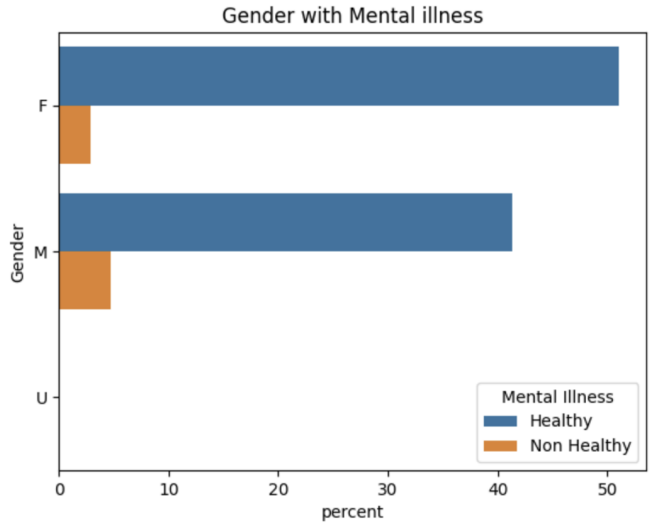
(vi) Distribution of Mental Illness by Age Group:



Orange bars represents those diagnosed with mental illness. We could insight like most hospital patients are older regardless of mental health status. Mental health cases appear more frequent in age groups 3 and 4 such as 30-49 and 50-69 respectively.

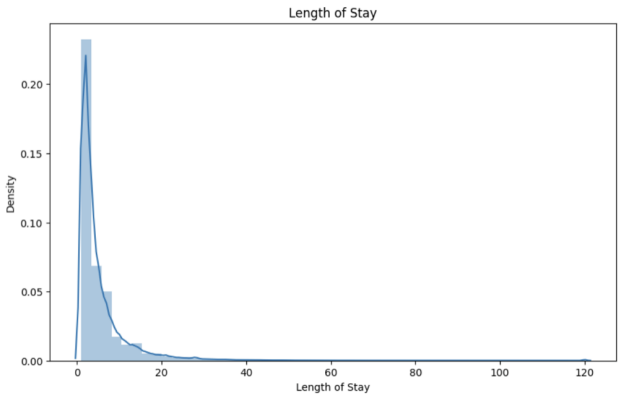
This helps us to understand that which age group are affected by mental illness the most. And confirm that working age adults from 18-69 are most vulnerable.

(vii) Distribution of mental illness via gender:



Orange bar slightly longer for males than females here suggesting a higher proportion of mental illness was diagnoses among male.

(viii) Distribution of Length of Hospital Stay:

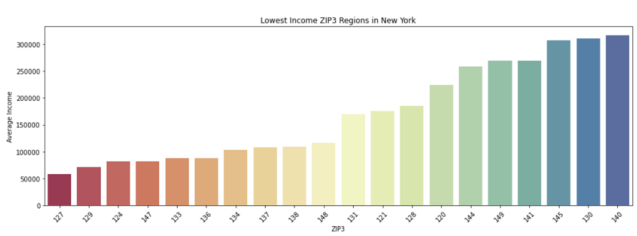


Here, we could see most patient stayed for short period of time around 1-5 days however, the graph skewed to the right meaning longer stays are but possible like patient stayed up to 100+ days. It could be that mental health hospitalizations might resolved within a few days to a particular hospital and admitted or recommend to another level of hospital.

D. Income dataset

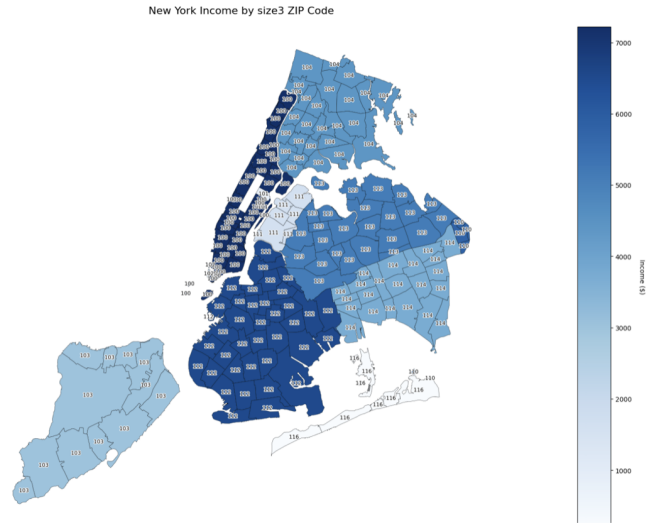
Income dataset consist of 12346 rows and 163 columns with detailed income related attributes. ZIP codes in original dataset were recorded in standard 5-digit format, but to perform regional aggregation and merge with other dataset, this were converted to the first 2 digits ZIP code known as ZIP3.

(i) Ranking of New York



X-axis shows the ZIP3 prefixes while Y-axis shows average income in dollars. Colored by ZIP3 with gradient from lowest to highest income regions. ZIP3 like 127, 129 have the lowest average incomes under appox. 90,0000. **Cross Dataset EDA-income vs hospitalization analysis:**

This was created merging dataset grouped by ZIP3, and visualize how income varies across regions where hospitalizations occurs. Color gradient- lighter shades represent lower income and darker shades with higher income.



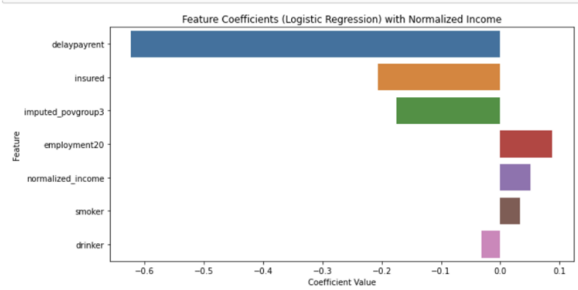
Lighter areas like 104,116 or 114 represent lower-income ZIP3 example Bronx. This could as well shows higher hospital or mental health burden. Darker areas like 100,112,110 like Manhattan and western Broklyn represent wealthier ZIP3. This map directly visualize the economic side of the relationship showing financial disparities exist.

V. Modeling and Evaluation

Using the processed and visualized dataset of survey dataset, we developed predictive model to identify if individuals at risk of poor mental health based on socioeconomic features. The target variable was based on low mood scores while predictor features include socioeconomic indicators such as income group, insurance, and so on and financial stress like delayed rent payments. Three machine learning models like **Logistic Regression, Random Forest and XGBoost** were performed to implement to classify individuals as At Risk or Not At Risk. Model performance was done using key evaluation metrics such as precision, recall, F1 score and confusion matrices. Among the models Logistic Regression performed best with F1 score of 0.66 with XGBoost closely following with 0.65 indicating string models.

A. Feature Coefficient- Logistic Regression

This show how each input variable affects the prediction of mental health risk both in positive or negative influence and in strength of that influence. Here, Negative coefficients to the left 0, decrease the likelihood of being classified as “At Risk”. Positive coefficient to the right of 0 increases the likelihood of being “At Risk”.

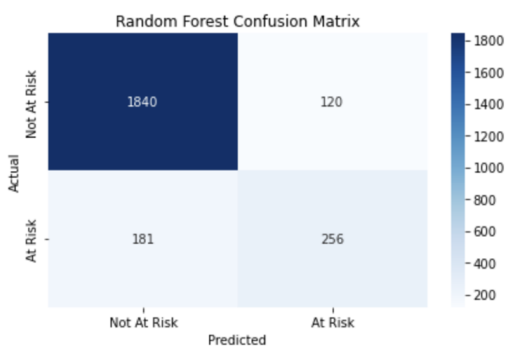


	LogisticRegression Coefficient
delaypayrent	-0.623051
insured	-0.206644
imputed_povgroup3	-0.175102
employment20	0.088038
normalized_income	0.051072
smoker	0.033787
drinker	-0.031330

Delaypayrent with coefficient -0.623 shows string negative impact like individuals who cannot pay rent on time are much likely to be classified as "At Risk".employment20 with +0.088 might shows lower income groups correlate with increased risk and smoker/drinker shows small coefficients with minor influence on classification. Through this we would see the transparency to the model of how much each variable contributes. This helps interpret which variable matter most and focus on supports policy.

B. Confusion Matrix+ F1 Score Comparison

A random forest confusion matrix was done to assess prediction quality, and reveals if the model is biased with imbalanced datasets. The comparison of F1 scores for two models -Logistic Regression+ Random Forest was done to evaluate which model performs better.



	Logistic_F1	RandomForest_F1
0	0.660601	0.629766

In confusion Matrix(Random Forest)

	Predicted: Not At Risk	Predicted: At Risk
Actual: Not At Risk	1840(True Negative)	120(False Positive)
Actual: At Risk	181(False Negative)	256(True Positive)

256 with True positive individuals with mental health risk correctly identified, 1840 True Negatives individuals healthy individuals correctly predicted. 120 False Positives predicted at risk, but they weren't and 181 False Negatives missed real at risk cases.

We tried with another model performance and evaluated F1 score but logistic Regression performs better than Random Forest.

This overall, confirms that ML can identify people at mental health risk. **Cross-Dataset Modeling** Due to the larger size of hospital admission dataset, we performed aggregration on mental health related admission by ZIP3 afterwards the resulting summary was merged with regional income data to perform a cross-dataset analysis of socioeconomic influence in order to understand how income disparities correlate with pattern in hospital utilization and patient outcomes on mental health of individuals.

We successfully merged hospital admission data and other dataset through ZIP3 creating comprehensive cross-dataset.

```
merged.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 49 entries, 0 to 48
Data columns (total 13 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   Zip Code - 3 digits             49 non-null     int64
 1   admission_count                 49 non-null     int64
 2   F                               49 non-null     int64
 3   M                               49 non-null     int64
 4   U                               49 non-null     int64
 5   total                           49 non-null     int64
 6   pct_female                      49 non-null     float64
 7   pct_male                        49 non-null     float64
 8   Total Costs                     49 non-null     float64
 9   Length of Stay                  49 non-null     float64
 10  index                           49 non-null     int64
 11  grouped_income_per_individual   49 non-null     float64
 12  mental_admission_count          49 non-null     int64
dtypes: float64(5), int64(8)
memory usage: 5.1 KB
```

```
                            OLS Regression Results
==================================================================
=======
Dep. Variable:        admission_count   R-squared:
0.151
Model:                            OLS   Adj. R-squared:
0.133
Method:                 Least Squares   F-statistic:
8.342
Date:                Mon, 19 May 2025   Prob (F-statistic):
0.00584
Time:                        14:32:32   Log-Likelihood:
-595.58
No. Observations:                  49   AIC:
1195.
Df Residuals:                      47   BIC:
1199.
Df Model:                           1
Covariance Type:            nonrobust
==================================================================
=========================
                                 coef    std err          t      P>|
t|      [0.025      0.975]
------------------------------------------------------------------
-------------------------
const                        1.984e+04   9665.331      2.052      0.0
46     392.794    3.93e+04
grouped_income_per_individual    13.1719      4.560      2.888      0.0
06       3.997      22.346
==================================================================
=======
Omnibus:                       54.841   Durbin-Watson:
1.586

Omnibus:                       54.841   Durbin-Watson:
1.586
Prob(Omnibus):                  0.000   Jarque-Bera (JB):
269.891
Skew:                           3.016   Prob(JB):
2.48e-59
Kurtosis:                      12.788   Cond. No.
3.06e+03
==================================================================
=======

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is
correctly specified.
[2] The condition number is large, 3.06e+03. This might indicate that t
here are
strong multicollinearity or other numerical problems.
```

Yes , Coefficient = 13.17: For every 1-unit increase in income, admissions increase by **~13.17** units, holding other factors constant. And it's statistically significant p= 0.006 (p< 0.050)

```
                            OLS Regression Results
==================================================================
===========
Dep. Variable:     mental_admission_count   R-squared:
0.152
Model:                                OLS   Adj. R-squared:
0.134
Method:                     Least Squares   F-statistic:
8.445
Date:                    Mon, 19 May 2025   Prob (F-statistic):
0.00557
Time:                            14:59:37   Log-Likelihood:
-472.31
No. Observations:                      49   AIC:
948.6
Df Residuals:                          47   BIC:
952.4
Df Model:                               1
Covariance Type:                nonrobust
==================================================================
=========================
t|      [0.025      0.975]
------------------------------------------------------------------
-------------------------
const                        1432.1634    781.064      1.834      0.0
73    -139.135    3003.462
grouped_income_per_individual     1.0710      0.369      2.906      0.0
06       0.330       1.812
==================================================================
=======
Omnibus:                       58.584   Durbin-Watson:
1.826
Prob(Omnibus):                  0.000   Jarque-Bera (JB):
327.195
Skew:                           3.222   Prob(JB):
8.92e-72
Kurtosis:                      13.897   Cond. No.
3.06e+03
==================================================================
=======

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is
correctly specified.
[2] The condition number is large, 3.06e+03. This might indicate that t
here are
strong multicollinearity or other numerical problems.
```

1) Effect of Income on Hospital Admission Rate- Linear Regression : In here, we tried to examine the relationship between Independent Variable grouped_income_per_individual (average income per ZIP3).

through the examine we could see:

Coefficient(13.17)- With every 1 unit increase in income, the number of admission to hospital increases by 13.17 with p-value 0.006 that's is less than 0.05 means statistically significant. With R-squared(0.151) explains income alone is nearly 15.1% of variable for hospital admission.

Though we might expect lower income relate to high admission to hospital, but the model suggested as income increases, hospital admission also rise, might be because of better access to healthcare, or more frequent elective admissions or higher population density in high income ZIP codes.

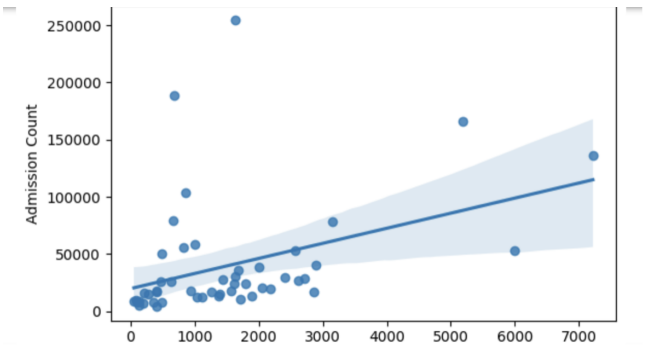
2) Does Income predict mental health hospitalization- ZIP3 level regression?: The model explored main concern if income levels are directly relate to number of mental health-related hospital admissions across different ZIP code.

The coefficient of 1.07 shows for every 1 unit increase in income mental health admissions increase by 1 unit with p-value 0.006 statistically significant and r-square of 0.152 explain 15.2% of variation for hospital admissions is by income alone that has an effect but limited.

The income does plays a role in mental health related admission.

3) Linear Regression Plot of Income vs Hospital Admission Count: : This shows relationship between income levels-ZIP3 and the number of hospital admission. The slop of the regression highlights as a positive correlation, as income increases hospital admission rise too. This finding aligns with previous analysis, where the coefficient for income was both positive and statistically significant (p = 0.006). Although the data points are somewhat scattered, the trend line, coupled with a shaded confidence interval, demonstrates that income can moderately predict admission counts; the R-squared value of approximately 0.13 suggests that income accounts for about 13% of the variation in hospital admissions. Interestingly, some ZIP3 regions with lower or moderate income levels still show high admission counts, implying that other factors—such as age, gender, and insurance coverage—also play a significant role. Additionally, there is variability in admission rates at

higher income levels, reinforcing the idea that income alone does not solely drive hospital admissions.

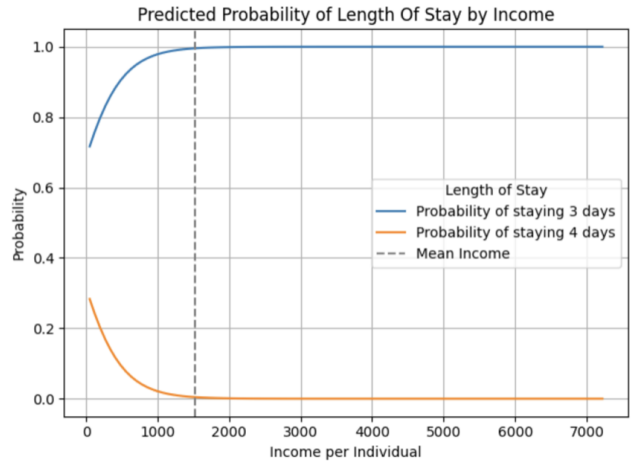


This visualization supports the goal of examining the association between economic factors, like income, and hospital admissions, highlighting that higher income areas often experience increased hospital admissions, potentially due to better access to healthcare, greater health awareness, or a higher burden of chronic diseases among insured populations.

4) Does Income predict length of stay in hospital- LR: The Ordered Logistic Regression study determines if income effects the length of hospital stays. The primary findings reveal a coefficient of -0.0031 for income, indicating a modest negative link between higher income and shorter stay resulting is not statistically significant (p-value = 0.152). The confidence interval [-0.007, 0.001] spans zero, showing that wealth has no significant influence on stay time.

```
Optimization terminated successfully.
         Current function value: 0.171993
         Iterations: 43
         Function evaluations: 80
                             OrderedModel Results
==============================================================================
Dep. Variable:         Length of Stay   Log-Likelihood:                -8.4276
Model:                   OrderedModel   AIC:                             20.86
Method:             Maximum Likelihood   BIC:                             24.64
Date:                  Mon, 19 May 2025
Time:                          15:30:19
No. Observations:                   49
Df Residuals:                       47
Df Model:                            1
==========================================================================================
                                    coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------------------
grouped_income_per_individual    -0.0031      0.002     -1.431      0.152      -0.007       0.001
3.0/4.0                           0.7732      0.950      0.814      0.416      -1.088       2.635
==========================================================================================
```

While higher income is associated with more hospital admissions, it does not seem to affect the length of stay significantly. This may be because individuals with higher incomes often have better access to preventive care and may rely on outpatient services rather than extended hospital stays. In contrast, lower-income individuals might delay seeking treatment, leading to shorter stays due to various constraints, such as discharge protocols and insurance limitations. Thus, the results clarify that while income affects admission rates, it does not significantly impact how long patients remain in the hospital.



5) Probability of Length of Stay at hospital by Income : The plot shows the inverse relationship between income and longer hospital stay and even if the relationship isnot statically significant, we can search a direction of further research. The current dataset is insufficient to confirm with high confidence but with more granular dataset may validate potential relationship.

VI. Discussion

This study presents a comprehensive approach to analyzing mental health outcomes using integrated survey, hospitalization, and income data from New York City. The results reinforce the importance of behavioral and environmental indicators in predicting mental distress and provide new insights into regional hospitalization trends for mental illness.

The classification modeling process showed that logistic regression outperformed other algorithms, producing an F1 score of 0.66. Key predictors of mental health risk included delayed rent payments, unemployment, and mood-related indicators derived from the Kessler Psychological Distress Scale (K6). These findings are consistent with prior research suggesting that financial strain and emotional distress are closely linked.

Interestingly, the cross-dataset regression revealed a positive correlation between income and the number of hospital admissions for mental illness, a result that may initially seem counterintuitive. One possible interpretation is that higher-income ZIP code regions may have better access to healthcare facilities and greater awareness of mental health conditions, leading to higher reported admission rates. Alternatively, the pattern may reflect the greater population density or healthcare utilization behaviors in certain urban neighborhoods.

However, the study found no statistically significant relationship between income and the length of hospital stays for mental health cases. This suggests that while income may influence access to care, it does not necessarily affect treatment duration once a patient is admitted.

The integration of the K6 scale into the target variable provided clinical grounding to the model's output and improved label reliability. This adds credibility to the machine learning

predictions and aligns the project more closely with real-world mental health screening practices.

Although the results offer meaningful insights, the study has limitations. ZIP3 aggregation, while useful for regional analysis, may obscure more localized disparities in care. Additionally, the survey data is self-reported, which can introduce bias or underreporting. Future work could explore more granular spatial data or incorporate real-time social media and mobility data to enhance temporal and behavioral tracking.

Overall, the findings support the value of combining behavioral and clinical datasets for mental health prediction and resource planning. They also highlight the need for targeted intervention strategies in regions with high risk but limited access to care.

VII. Conculsion

This study explored the relationship between financial conditions and mental health outcomes in New York City through a multi-dimensional data mining approach. By integrating survey-based behavioral data, hospital admission records, and regional income levels, we were able to identify patterns of psychological distress and healthcare utilization across ZIP3 regions.

Using machine learning classification models—specifically Logistic Regression, Random Forest, and XGBoost—we predicted mental health risk based on individual-level features such as mood scores, alcohol consumption, income group, and rent payment status. Among these models, Logistic Regression achieved the best performance, with an F1 score of 0.66, and highlighted socioeconomic stressors like rent delays and employment status as key predictors of mental health vulnerability.

The Kessler Psychological Distress Scale (K6) was used to support the construction of the target variable for risk classification, improving the validity of our mental health risk labeling. In addition to classification, regression analysis showed that income had a modest yet statistically significant positive correlation with the number of mental health-related hospital admissions, possibly due to better access to care in higher-income ZIP codes. However, no significant relationship was found between income and the length of hospital stays, suggesting that economic status does not impact treatment duration once admitted.

Overall, the study demonstrates that economic and behavioral indicators can effectively inform mental health risk prediction and help identify regions with disproportionately high hospitalization rates. These insights may assist policymakers and public health agencies in targeting mental health interventions and allocating resources more equitably across the city.

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